



NATIONAL OPEN UNIVERSITY OF NIGERIA

University Village, Plot 91, Jabi Cadastral Zone, Nnamdi Azikiwe Expressway, Abuja

FACULTY OF SCIENCES

Computer Science Department

2022_2 EXAMINATION

Course Code: **CIT 308**

Course Title: **Formal Methods and Software Development**

Credit: **3 Units**

Time Allowed: **3 hours**

Instruction: **Answer Questions One (1) and any other THREE (3) questions**

Questions One (25 Marks) – Compulsory

- 1 (a) Outline any two (2) characteristics of high-quality software. **(2 marks)**
- 1 (b) Describe four (4) phases of a typical software development cycle. **(6 marks)**
- 1 (c) Critique two (2) classifications of formal methods. **(5 marks)**
- 1 (d) Complete the truth table for the proposition $(A \rightarrow B) \vee (B \rightarrow A)$

A	B	$(A \rightarrow B)$	$(B \rightarrow A)$	Result: $(A \rightarrow B) \vee (B \rightarrow A)$
T	T			
T	F			
F	T			
F	F			

(12 marks)

Question 2

2 (a) State the condition(s) under which a formula is:

- i) A tautology
- ii) Consistent
- iii) Inconsistent

(1 mark each – Total = 3 marks)

2 (b) For the statements below, identify the (i) constant (ii) variable object (iii) function –

An ostrich has wings can fly; An eagle has wings can fly.

(3 marks)

2 (c) Explain with examples the following terminologies:

- i. Family sets
- ii. Power sets
- iii. Disjoint sets

(3 mark each – Total = 9 marks)

Question 3

3 (a) Explain the terms (i) Bound variable (ii) Free variable.

(3 marks)

3 (b) Given the following:

$$A = \{1, 2, 5, 7, 9, 10\}$$

$$B = \{-1, 3, 5, 8, 10\}$$

$$C = \{2, 4, 6, 8, 10\}$$

Calculate the following:

i. $A \cup B$	v. $A \cap B$
ii. $A \cup C$	vi. $A \cap C$
iii. $B \cup C$	vii. $B \cap C$
iv. $A \cup B \cup C$	viii. Show your results in a Venn diagram

(12 marks)

Question 4

- 4 (a) Describe with examples two types of sequences. (3 marks)
- 4 (b) (i) State the formula for a geometric sequence.
(ii) Given: $X = 4, 16, 64, 256, 1024$, Calculate the 10th term. (3 marks)
- 4 (c) List and explain with examples three (3) types of proofing methods. (9 marks)

Question 5

- 5 (a) Describe any five (5) stages (or activities) in the *Software Development Life Cycle (SDLC)* (5 marks)
- 5 (b) Outline any four (4) processes that *formal specification* follows. (4 marks)
- 5 (c) Discuss any three (3) types of the testing concept - *Test Size* - in software development. (6 marks)

Question 6

- 6 (a) Discuss any two (2) laws of *Software Evolution*. (4 marks)
- 6 (b) Describe any two (2) results of software design levels. (4 marks)
- 6 (c) (i) Explain the term Cohesion. (2 marks)
(ii) Describe any five (5) types of cohesion (5 marks)